

COURSE INTRODUCTION

Lecture 1 (Introduction)

procedures for the definition of techniques for the design of Electrical drives
with usage of MATLAB/SIMULINK for simulation

EXAM ORAL exam (eng/ita) + report about an exercise activity
↳ 2 questions about dynamic model of some electrical machine

(organized by appointment, 10 days before) ⇒ also if during official session send anyway an email to program the examination

Example OLDCOGRAPHIC Ventilation

using an electrical drives subjected to disturbances → to reject the disturbance you have to

carefully on the BANDWIDTH ← design the CONTROL LOOP appropriately for the type of disturbing
feedback given on the system?

the control of the system is given by a position sensor (by speed integration? NO)

a cheaper solution to measure the position, instead of a complex sensor you can use another solution → absolute position of the system by HALL EFFECT sensor

that give the permanent magnet position correctly. → using the permanent magnet on the rotor to obtain a correct position measurements

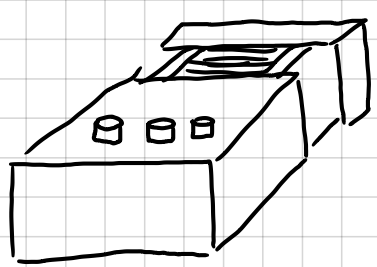
(otherwise it can be used an ENCODER)

disturbing the permanent magnet the system acts badly → if using also an encoder (with good step precision you can increase control performances on the DC machine and change led configuration)

↓ this is a system with a POWER supply directly connected, with constant speed of Rotor, NO power control



Example AC power supply

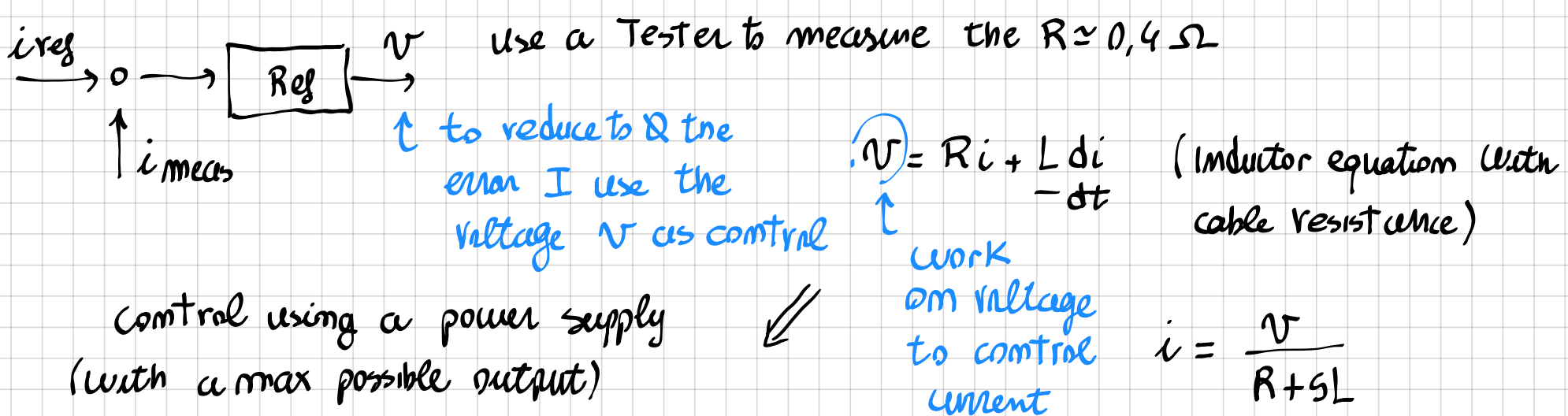


give you info about nominal current → the value of current that allows the device to have a long life-time

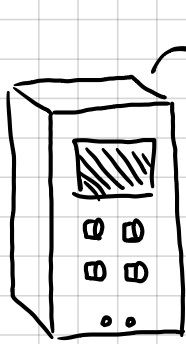
if you know the system is cheaper an OPEN LOOP simple approach without sensors

↳ but with OPEN LOOP there is NO feedback, risky if you need precision

Open Loop if I want to power supply it correctly I may use a tester to measure resistance (if closed loop I need to measure the current feedback)

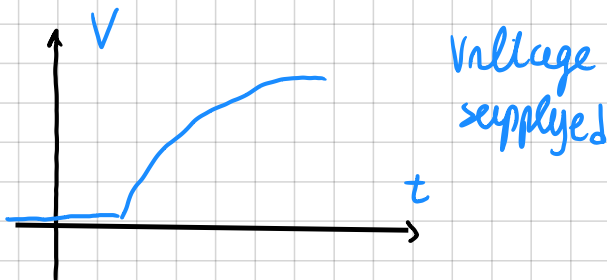


control using a power supply (with a max possible output)



connected to an oscilloscope

OPEN LOOP approach



Looking this you can infer the time constant $\tau = L/R$ of the system

(the open loop can help the closed loop) ← feed forward using syst knowledge

IF you want to reduce the time constant τ , you increase the voltage

how to reduce the transient time... by an

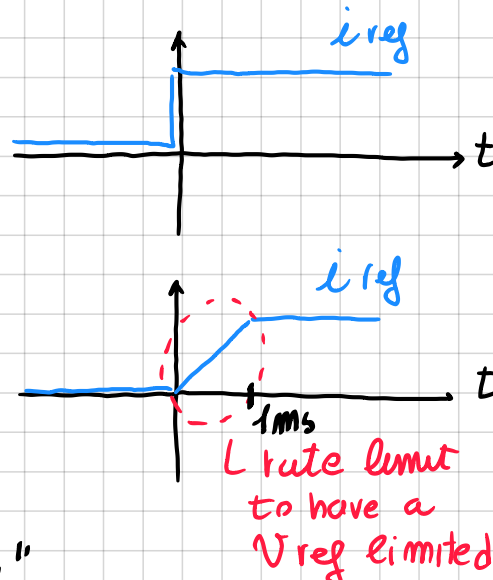
increase of R with an external Resistance ⇒ here the problem is caused by the open loop approach

from

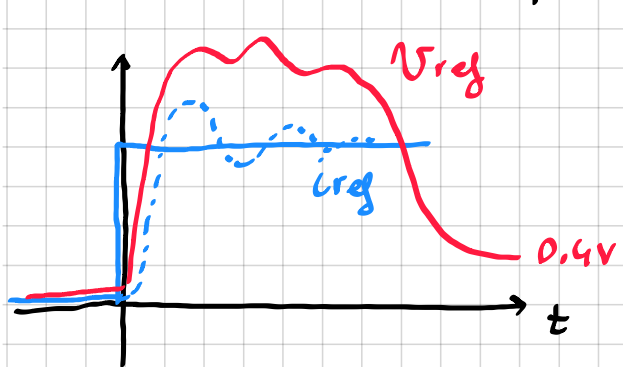
$$V = Ri + L \frac{di}{dt} \Rightarrow V_{reg} = Ri_{reg} + L \frac{di_{reg}}{dt}$$

performance limitation due to open loop control

$$V_{reg}(0) = 4V + 0.4 = 4.4V$$



while on Closed Loop



PI control

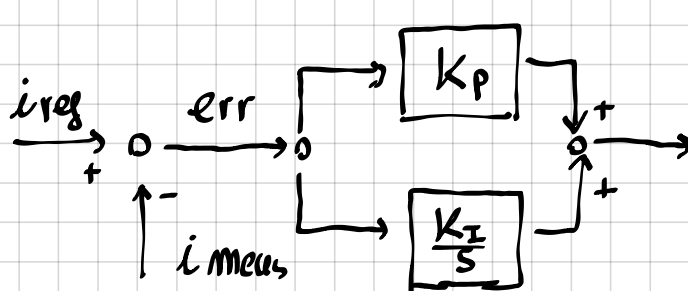
(not D because "derivative"

is hard to obtain) Proportional + Integrational control

design of controller made on

low error → if high error

bad design → SATURATION



Using an ANTI-WIND UP (to solve saturation issue)

manage the saturation on the system

Similar approach on MACHINE, with also torque acting on the system

You can use { open loop close loop } help close loop with parallel feed forward approach

Information about the course

email: francesco.castellidezza@polimi.it

phone: 02 2399 8455

Office: Building B23 (Bovisa Campus), DMEC,
Via La Masa 1, Milan
Building 7 (Leonardo Campus), DEIB,
Prof. Piegari (4125), Piazza L. da Vinci 32, Milan

Lecture timetable:

- Monday, 15.15-18.15** – Lesson (T.1.3, 300)
- Tuesday, 16.15-19.15** – Exercise/Lab (8.0.1, 260)
- Wednesday, 8.15-10.15** – Lesson (T.2.2, 312)

Books: - Lesson notes on WeBeep website and on castellidezza.faculty.polimi.it
- N. Mohan, T.M. Undeland, W.P. Robbins, «Power Electronics:
Converters, Applications, and Design», Editore: John Wiley & Sons

(Consultation timetable: by appointment (Leonardo or Bovisa)
campus)



Goal: define procedure to design an electrical drives
↓ composed by

↙ electrical machine, the power supply electronics and
control system

to do this is

necessary knowledge on DC, Induction, synchronous machine ecc...
+ power electronics in order to supply
+ control system

modelling of
machine → dym

↔ equation to find the expression of the **torque**

(to find it.. express T to control the mechanical load, so
the output effect)

Table of Contents

Lectures program

- **Principle** of electrical engineering and energy conversion
- **DC motor**: steady-state and dynamic model, static converter, flux, speed and torque control schemes.
- **Induction motor**: steady-state equivalent circuits; 4 and 5 parameters dynamic model, voltage and current source inverters, state observers, field oriented control, scalar control scheme, DSC and DTC.
- **AC Brushless**: dynamic model, field oriented control. Synchronous Reluctance machine.
- **DC Brushless**: static converter and control scheme.
- **Switched reluctance machine (SRM)**: static converter and control scheme.
- Industrial Case studies

Computer laboratory and experimental activities: → simulation on MATLAB, SIMULINK

- Design of the control of a **dc machine**
- Design of the **field oriented control** of an induction machine
- Design of the **scalar control** of an induction machine
- Design of the **direct torque control** of an **induction machine**
- Design of the **field oriented control** of an **ac brushless**

during exercise activity

Course assessment

The **exam** consists of an **oral test** (in Italian or English) to verify the knowledge of the electrical drives and a discussion on a **report of one of the exercise activities, taken during the semester.**

↑ report one of exercise activity during exam (send it before)

The exam consists of **two main questions on two different electrical drives** in order to test the knowledge of the student about:

- **dynamic models** of the electrical machine, starting from its structure
- **expression of the electromagnetic torque**
- **control schemes** (mainly torque and flux control loops)
- **operating regions**
- **estimators** (eventually)

The **report** has to show the definition of the parameters of the control system and the results of the simulation of the application (Matlab/Simulink).

↑ show result of a problem on PDF form



by appointment, ask for an appointment **10 days before** to schedule (also on official session dates)

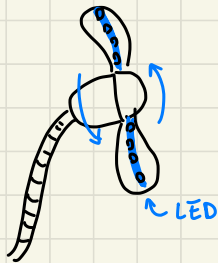
+ **Tutoring activity** (time table on weebep)

Symbols (Notation)

Abbreviation	Variable	Unit
v	voltage	V
i	current	A
$e(t)$	induced voltage e.m.f.	V
R	resistance	Ω
L / C	inductance/capacitance	H / F
R / A	reluctance/permeance	H^{-1} / H
ϕ	magnetic flux	Wb
ψ	flux linkage	Wb
f	force	N
T	torque	Nm
v	linear speed	m/s
ω	angular speed/frequency	rad/s
J	inertia	$kg \cdot m^2$
n_p	number of pole pairs	

⇒ **Example** ideas about automation/control system on electrical device

• **holographic Vent** simple electrical drives



Trying to **disturb** this system, it does NOT remain stable...

the whites with led are NOT stable when disturb by hand the position!

disturbance even if apply a step!

perturbed from steady state

Bad dynamical performance

⇒ bad control response time, of the system has a

system is NOT enough responsive
bad performance when a step acts on system,

NOT ROBUST

← design control bandwidth bad for low variation command

↳ design the controller on the disturbance type

long bandwidth is cost! $\Rightarrow$ to increase BW you need high performance controller

the BW of control scheme is influenced also by the fact of the syst. scheme

↓
problem on $\rightarrow$ CONTROL (adaptive control if you want good perform.)

(system is as it is, more changeable) $\rightarrow$ think about the feedback
given by the control of the POSITION (sensors)
(speed feedback is not enough, and integration of speed has slipped, NOT good)

↓
Cheap measurement of position?

(We have an open loop system) $\rightarrow$ no response if you perturb the syst.
also on printer you use mechanical switches to have info about position

here you use an Hall effect sensor $\Rightarrow$ permanent magnet position, information 0/1 on magnet on ROTOR to check
(while using an ENCODER you get the actual position)
When magnet on sensor $\rightarrow$ absolute sensor

here one hall effect sensor $\rightarrow$ when disturbing the system is not robust, don't correct until next rotation

$\hookrightarrow$ when magnet reposition

to increase performance I can introduce an Encoder (with its resolution)

$\uparrow$ DC machine with a cheap control syst using info of hall effect sensor, with a led configuration changing @ each position $\rightarrow$

during 360° interpolate, forecast position of lighting led

knowing Rpm $\rightarrow$ forecast next position

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↓
changing speed is NOT necessary, if changing the speed by disturb, is robust against speed variation

⇓
suppose you have some instruments

- power supply
- tester
- a new device to study



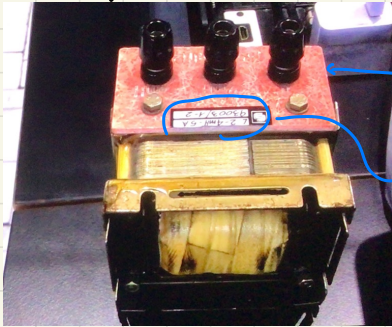
Example 2

- tester
- cables
- power supply
- oscilloscope
- new device

PROBLEM

Goal control the current
↳ inside the device

↓
to obtain 2A of current inside



this is a **INDUCTOR**

3 terminals

L. 2-4 mH - 5A
93003 / 1-2

nominal
current

↓ ensure that
working continuously
on this current,
the life time
is long!

(control a current equal to 2A?)

INDUCTOR

how to realize a
current of 2A?

↳ how to correctly use the given
instruments to obtain our goal

(close/open loop approach?)



if you know the syst, you can use
open loop, cheaper solution without

↙ sensor

JUST using a power supply

o BUT in open loop

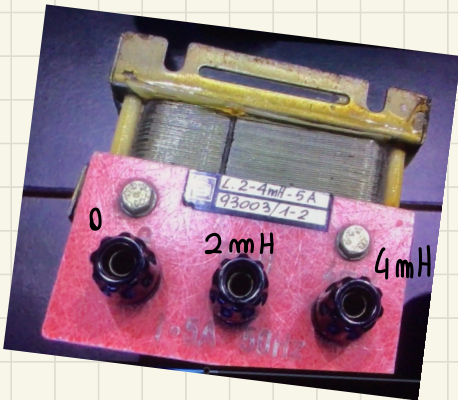
+ you have no feedback + problem related to parameters of syst.

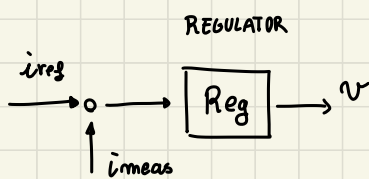
which changes, **R(Temperature)** there are
additional info needed

+ how to be sure

that input is precise ⇒ you need to use a tester and
measure resistance

(while on close loop you measure amount to be sure of output)





to reduce to 0 the $i_{ref} - i_{meas}$ I act on the voltage which influence the current i as:

↓ inductor dynamic (real syst)

work on v to act on i

$$V = Ri + L \frac{di}{dt}$$

one pole T.F (on steady state)
 $V = Ri$
 all constant

$$i = \frac{V}{R + sL}$$

With a TESTER measure R on steady state
 $0.4 \Omega = R$

($V = Ri$ dym. steady) (you can also supply the syst in another way...)



use power supply to give a max current of 2A measuring in a close loop approach a bad measure

(good procedure by amperometer + voltmeter)

⚡ (if supply with 0.5V, measure 4.06A)

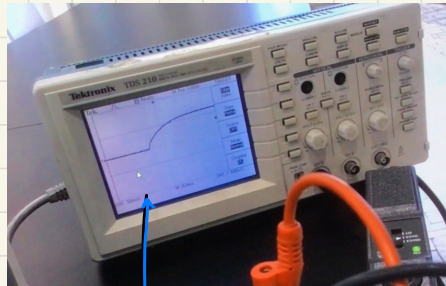
oscilloscope

↳ with the tester you measure the current

transient...

(I order system)

on open loop approach!
 you can introduce v



measured by knowing τ, R

$$\tau = (L/R)$$

time constant

reach steady state in 5τ

how to reduce τ on open loop? $\Rightarrow$ (sometimes used together closed loop)

(introducing R_{ext} in series you start a new transient, increase time...)

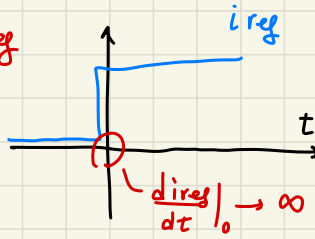
(feed forward when using open loop) + closed loop

the problem on τ is due to the open loop approach $\Rightarrow$

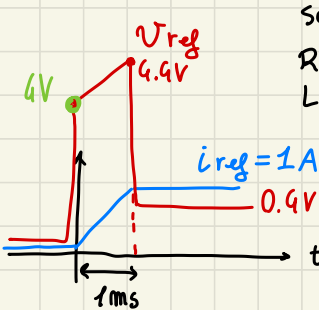
from dym eq. $V_{ref} = R i_{ref} + L \frac{d i_{ref}}{dt}$ on open loop approach

but this $\downarrow$ Use the knowledge of overall syst.
cannot be realized, if step on i_{ref}

on reality you don't use step, but introduce a Rate limiter on i_{ref}



No possible ∞ power for instantaneous change on V



Suppose
 $R = 0.4 \Omega$
 $L = 4mH$

(open loop)

$\bullet V_{ref}(t=0) = 0 + 4mH \frac{1}{1ms} = 4V$

on steady state condition,
only V acting on R

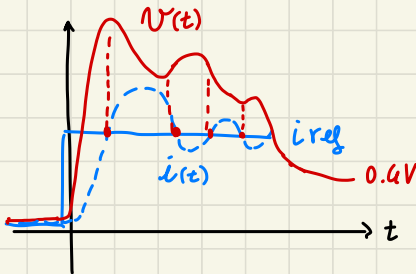
to have μs

means KV, too much voltage required $\rightarrow$ power supply limitation,
given by open loop approach

Closed loop

(input step)

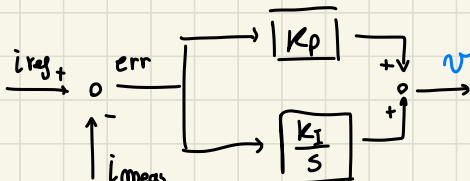
there is some oscillation on the control V
 V control to i follow i_{ref}



PI controller

(derivative part amplify the noise $\sim$ times, hard to use)

$\downarrow$
so we use PI
control (proport + integration)



if you want high
bandwidth V is asked to be
very high But power supply is
limited $\Rightarrow$ to reduce this problem
ANTI wind up

anti WIND UP $\Rightarrow$ PI controller stopped before reaching saturation point
otherwise if you request v higher $v \uparrow \downarrow$ continuously
 $\swarrow$ (OUTPUT saturation!)
takes time to store integration term

usually you often face the saturation problem $\Rightarrow$ manage it by
power supply limitation $\uparrow$ ANTI WIND-UP

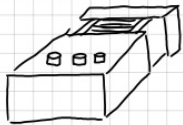
on real plant error high brings to saturation!

$\Downarrow$
if err $\uparrow$ for some you face SATURATION

$\uparrow$
use this idea for machines, acting on TORQUE

open
closed loop $\rightarrow$ helped with feed forward in parallel
to closed loop to correctly reach the
desired reference

Example AC power supply



↓ give you info about nominal current → the value of current that allows the device to have a long life-time

if you know the system is cheaper an OPEN LOOP simple approach (without sensors)

↳ but with OPEN LOOP there is NO feedback, risky if you need precision

Open Loop if I want to power supply it correctly I may use a tester to measure resistance (if closed loop I need to measure the current feedback)

$i_{ref} \rightarrow 0 \rightarrow$ [Reg] $\rightarrow V$ use a Tester to measure the $R \approx 0,4 \Omega$

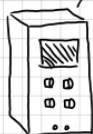
↑ i_{meas}
↑ to reduce to R the error I use the voltage V as control

$$V = Ri + L \frac{di}{dt} \quad (\text{Inductor equation with cable resistance})$$

work on voltage to control current

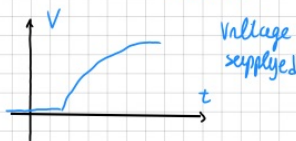
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how to reduce the transient time... by an

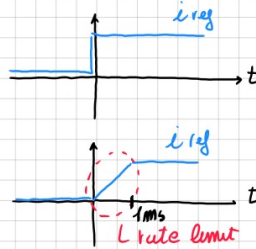
increase of R with an external Resistance → here the problem is caused by the open loop approach

from

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performance
limitation due
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$$V_{reg}(s) = 4V + 0.4 = 4.4V$$



$$R = 0.4 \Omega$$

$$L = 4mH$$

while on closed loop



PI control

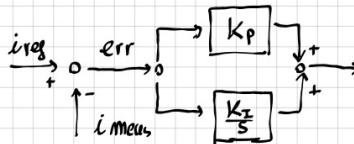
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low error $\rightarrow$ if high error

bad design $\rightarrow$ SATURATION



using an ANTI-WIND UP
(to solve saturation issue)

manage the saturation
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similar approach on MACHINE, with also torque acting on the system

You can use $\begin{cases} \text{open loop} \\ \text{close loop} \end{cases}$ $\rightarrow$ help close loop with parallel feed forward approach